

2. (3 points)

Given the following information as of December 31, 2018:

Accident Year	Earned Premium (\$000)	Incremental Paid Loss (\$000) as of (months)		
		12	24	36
2016	5,000	1,800	700	500
2017	6,000	2,000	800	
2018	8,000	2,200		

- Assume there is no further development after 36 months.
- $Var[U_i] = Var[U_i^{BC}]$

a. (1.5 points)

Calculate the accident year 2018 Benktander reserve estimate (R^{GB}).

b. (0.5 point)

Calculate the accident year 2018 optimal credible reserve estimate (R_c).

c. (0.5 point)

Identify which of R_c or R^{GB} is the preferable reserve from a statistical point of view and briefly describe a supporting reason.

d. (0.5 point)

Describe the effect on the Benktander credibility for accident year 2018 if the incremental paid loss from 12 to 24 months for accident year 2017 was greater than the value in the table above.

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QUESTION 2	
TOTAL POINT VALUE: 3 points	LEARNING OBJECTIVE(S): A1
SAMPLE ANSWERS	
Part a: 1.5 points	
<u>Sample 1</u> $m_1 = \frac{1800+2000+2200}{5000+6000+8000} = 0.316$ $m_2 = \frac{700+800}{5000+6000} = 0.136$ $m_3 = \frac{500}{5000} = 0.1$ $ELR = 0.316 + 0.136 + 0.1 = 0.552$ $p_1 = 0.316/0.552 = 0.572 \quad p_2 = (0.316 + 0.136)/(0.552) = 0.819 \quad p_3 = 1$ $R^{GB} = q_k * U^{BF}$ $U^{BF} = 2200 + (1 - 0.572) * 0.552 * 8000 = 4090$ $R^{GB} = (1 - 0.572) * 4090 = 1750.52$ <u>Sample 2</u> $m_1 = \frac{1800+2000+2200}{5000+6000+8000} = 0.3158$ $m_2 = \frac{700+800}{5000+6000} = 0.1364$ $m_3 = \frac{500}{5000} = 0.1$ $ELR = \sum m_i = 0.5522$ $p_1 = m_1/ELR = 0.5719 \quad q_1 = 0.4281 \quad \rightarrow 2018$ 2018: $R^{ind} = \frac{Loss}{p_i} * q_i = \frac{2200}{0.5719} * 0.4281 = 1646.67$ $R^{Coll} = EP * ELR * q_i = 8000 * 0.5522 * (0.4281) = 1890.91$ $Z^{GB} = p_i = 0.5719$ $R^{GB} = Z^{GB} * R^{ind} + (1 - Z^{GB}) * R^{Coll} = (0.5719) * 1646.67 + (1 - 0.5719) * 1890.91 = 1751.22 \text{ (in 000s)}$ <u>Sample 3</u> $m_1 = (1800 + 2000 + 2200)/(5000 + 6000 + 8000) = 0.316$ $m_2 = (700 + 800)/(5000 + 6000) = 0.136$ $m_3 = 500/5000 = 0.1$ $ELR = \sum m_i = 0.552$ $p_1 = 57.2\% \quad p_2 = 81.9\%$ $R^{CL} = \frac{(2200)}{57.2\%} * (1 - 57.2\%) = 1646$ $R^{BF} = 8000 * (0.136 + 0.1) = 1888$ $Z = p_1 = 0.572$ $R^{GB} = (0.572) * 1646 + (1 - 0.572) * 1888 = 1750$	

Sample 4

$$ULT^{GB} = CL Ult (1 - q^2) + U_0 * (q^2)$$

$$m_1 = \frac{1800+2000+2200}{5000+6000+8000} = 0.316$$

$$m_2 = \frac{700+800}{5000+6000} = 0.136$$

$$m_3 = \frac{500}{5000} = 0.1$$

$$ELR = \sum m = 0.552$$

$$p_1 = 0.316/0.552 = 0.572 \quad q_1 = 1 - 0.572 = 0.428$$

$$AY 2018 R^{GB} = (1 - 0.428^2) * (2200) * \left(\frac{1}{0.572} - 1\right) + 0.428^2 * (0.552 * 8000 - 2200) = 1750$$

Sample 5

Since exposure has been increasing, I'm going to model loss ratio.

Cumulative Loss	12	24	36
2016	1,800	2500	3000
2017	2,000	2800	
2018	2,200		

$$12 - 24factor = \frac{(2500+2800)}{(1800+2000)} = 1.395 \quad 24 - 36factor = \frac{3000}{2500} = 1.395$$

$$12 - Ult = 1.395 * 1.2 = 1.674$$

$$CL Ult \text{ for AY 2018} = 2200 * 1.674 = 3682.8$$

Since we are not giving an expectation at losses, I'm going to use Hurlimann's approach to estimate loss ratio.

Loss Ratio

$$m_k = \frac{1800+2000+2200}{5000+6000+8000} + \frac{700+800}{5000+6000} + \frac{500}{5000} = 0.552$$

$$U_0 \text{ for AY 2018} = 8000 * 0.552 = 4416$$

$$\text{Paid \% for AY 2018} = \frac{1}{1.674} = 0.597$$

$$U^{BF} = 2200 + (1 - 0.597) * 4416 = 3979.65$$

$$U^{GB} = 2200 + (1 - 0.597) * 3979.65 = 3803.8$$

$$R^{GB} = 3803.8 - 2200 = 1603.80$$

Part b: 0.5 point

Sample 1

$$\text{Since } \text{Var}(U_i) = \text{Var}(U_i^{BC}) \quad Z^{opt} = \frac{p_i}{p_i + \sqrt{p_i}} \text{ because } t_i = \sqrt{p_i}$$

$$R_c = Z^{opt} * R^{Ind} + (1 - Z^{opt}) * R^{Coll}$$

$$R_c = 1646.67 * 0.4307 + 1890.91 * (1 - 0.4307) = 1785.74 \text{ (in 000s)}$$

Sample 2

$$Z^c = \frac{p_k}{p_k + \sqrt{p_k}} = 0.572 / (0.572 + \sqrt{0.572}) = 0.431$$

$$R^{ind} = \frac{2200}{0.572} - 2200 = 1646.15$$

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$$R_c = Z^c * R^{Ind} + (1 - Z^c) * R^{Coll} = 0.431 * 1646.15 + (1 - 0.431) * (4090 - 2200) = 1784.9$$

Sample 3

$$U^{CL} = \frac{2200}{57.2\%} = 3846$$

$$U^{BF} = 4090$$

$$Z = \frac{P}{P + \sqrt{P}} = 43.1\%$$

$$U_c = Z * U^{CL} + (1 - Z) * U^{BF} = 3988$$

$$R_c = 3985 - 2200 = 1785$$

Sample 4

$$CDF = 1.2 * 1.3947 = 1.67364 \quad p_i = \frac{1}{1.67364} = 59.75\%$$

$$Z = \frac{p_i}{p_i + \sqrt{p_i}} = 0.5975 / (0.5975 + \sqrt{0.5975}) = 0.436$$

$$\text{Res CL} = 2200 * (1.6736 - 1) = 1482.00$$

$$\text{Res BF} = (1 - 0.5975) * (0.552 * 8000) = 1777.44$$

$$\text{AY 2018 opt cred res} = 1482 * 0.436 + (1 - 0.436) * 1777.44 = 1648.63K$$

Part c: 0.5 point

Sample Responses

- R_c (optimal credibility) is a preferable reserve from a statistical stand point because it minimizes the MSE of the estimate, $MSE_c < MSE_{GB}$
- R_c is preferred because optimal credibility has the lowest of variance.
- R_c is the preferable reserve because it minimizes the MSE of the reserve.

Part d: 0.5 point

Sample Responses

- If that loss was greater, then the overall loss ratio would be larger. Since the incremental LR at 12 months stays the same, p would decrease, which means the credibility also decreases. If the incremental loss was greater than above, this would increase m_2 , which would also increase the ELR. m_1 would stay the same so $p_{2018} = m_1 / \text{ELR}$ would decrease. Since the credibility for Benktender is $Z = p_{2018}$ the credibility would decrease as well.
- If 12-24 incremental paid loss increased, this increase your 12-24 LDF. Since the % paid is based off of the calculated CDFs ($1/\text{CDF}$), the % paid at 12 months would decrease. Since your credibility factor is % paid, the credibility going to the current paid losses would decrease.
- So if it wasn't 800, if it was bigger:
 - o m_1 would remain the same
 - o m_2 would increase
 - o m_3 would remain the same
 - o $\text{ELR} = \sum m_k$'s would increase because m_2 is larger

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$\text{Cred } Z^{\text{GB}} = p_1 = \frac{m_1}{\text{ELR}} = \frac{m_1 \text{ remains the same}}{\text{ELR would increase}} \rightarrow p_1 \text{ would decrease so } Z \text{ decreases}$
 Answer: The credibility Z^{GB} would decrease

- ELR would increase and m_1 would be unaffected, leading to a decrease in p and a following decrease in Z ($Z^{\text{GB}} = p$). So credibility would decrease (more weight R^{Coll})

EXAMINER'S REPORT

Candidates were expected to know the key assumptions underlying the Benktander and Optimal reserve estimates, how to apply credibility, including mechanics of the methods (e.g., loss ratio based payout factors), be able to determine whether a particular reserve estimate is or is not preferable for the given condition, and understand the impact on Benktander credibility if loss condition has changed.

Part a

Candidates were expected to know how to calculate Benktander reserve estimate correctly, including all of the required components such as Hurlimann loss ratio payout factors, ELR and p (which is the Benktander credibility). They were expected to know how to calculate individual reserve estimate, collective reserve estimate, and application of credibility to these methods to get the Benktander reserve estimate.

Common mistakes included:

- Not using the right Benktander reserve estimate formulae
- Calculation error in the components in Benktander reserve estimate formulae
- Mistakenly considered BF reserve estimate as Benktander reserve estimate
- U^0 was incorrectly used instead of U^{Coll}
- Application of the wrong credibility (e.g., use $(1-p)$ as credibility on Individual reserves).

Part b

Candidates were expected to know how to calculate optimal credible reserve estimates correctly. They were also expected to be able to calculate optimal credibility given assumption in question and apply credibility correctly.

Common mistakes included:

- Not using the correct optimal credible reserve estimate formulae (e.g., R^{GB} was mistakenly used instead of R^{BF})
- Calculation error in the components in optimal credible reserve estimate formulae
- Application of the wrong credibility (e.g., use $(1-p)$) as credibility on Individual reserves

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Part c

Candidates were expected to know which reserve estimate (Benktander or optimal credible) is preferable and the reason of such conclusion, from a statistical point of view, given the assumption in the question.

Common mistakes included:

- Misstating that Benktander reserve estimate is preferable
- Not able to provide the rationale from statistical point of view
- Calculating MSE for Benktander and optimal credible reserve estimate with the wrong formula or calculation error

Part d

Candidates were expected to assess how the changes in incremental losses can impact Benktander credibility and provide the rationale for it.

Common mistakes included:

- Incorrect conclusion regarding the credibility impact
- Using the wrong credibility to base conclusion on
- Not providing any supporting reason for conclusion